Scalable Library Management System: Architecture, Implementation, and Performance Evaluation

Coursework: UCS3513 — System Design Laboratory | **Department:** CSE, SSN College of Engineering
Problem Statement: Problem Statement 3 — Scalable Library Management System (Athena LMS) | **Term:** V Semester

Executive Summary

Modern institutional libraries handle exponentially growing traffic encompassing thousands of concurrent student requests, tens of thousands of catalogue titles, physical copy inventory, circulation lifecycles, waitlists, and fine transactions. This project presents the architecture, complete full-stack implementation, formal verification test suite, and empirical performance benchmarks for **Athena LMS**, a distributed, horizontally scalable Library Management System. The system incorporates all eight curriculum-mandated components: Token Bucket Rate Limiting, Database Sharding, Consistent Hashing, Key-Value Storage, Snowflake 64-bit ID Generation, URL Shortening, Asynchronous Web Crawling, and Prefix Trie Autocomplete. Comprehensive multi-tier benchmarks demonstrate sustained throughput up to 504 req/sec, zero double-borrowing under 10-thread concurrency race conditions, and complete loan renewal lifecycle enforcement.

1. System Design Component Mapping & Implementation

Component	Library Functionality	Architectural Rationale	Benchmark Metric
1. Rate Limiter	Auth & Search Endpoints	Redis Lua Token Bucket (cap=20, refill=10/min) shields against DoS attacks.	**78.0% Throttled** under burst stress
2. Sharding	Members & Catalogue	Hashed on memberId (uniform distribution) & Range/Category on books.	Zero hotspots verified
3. Consistent Hashing	Microservice Routing	MD5 Ring with 160 vnodes per host minimizes cache churn during node scale.	**28.3% Churn** (vs Modulo 75.1%)
4. Key-Value Store	Catalogue Cache	Redis Cache-Aside read-through with write eviction and resilient fallback.	**46.4ms to 8.0ms** read latency
5. Unique ID Gen	Loans, Barcodes, Fines	Distributed 64-bit Snowflake algorithm: time-ordered, coordination-free.	**0 Collisions** / 50,000 IDs
6. URL Shortener	Item Sharing (/s/{code})	Bijective Base62 encoding engine with O(1) redirection lookup.	**21.4ms** Avg redirection latency
7. Web Crawler	Catalogue Ingestion	Asynchronous multi-threaded worker pool with ISBN deduplication.	**43.2 books/sec** throughput
8. Autocomplete	Search Bar Instant Hints	In-memory 26-way Prefix Trie indexing 37 titles and authors with popularity ranking.	**< 1.0ms** lookup latency

2. Concurrency Engine & Loan Lifecycle Architecture

Atomic Borrowing: To eliminate double-borrowing race conditions when multiple members request the last remaining physical copy simultaneously, the system avoids naive optimistic retry loops and executes an atomic state transition directly in MongoDB via findAndModify. A multi-threaded benchmark with 10 threads competing for 1 copy proved exactly 1 success (200 OK) and 9 conflicts (409).

Loan Renewal Lifecycle: Implements full renewal workflow enforcing four strict invariants: (1) Ownership verification, (2) Active loan status, (3) Maximum renewal limit capped at MAX_RENEWALS = 2, and (4) Waitlist Queue Defense —

renewals are immediately denied (HTTP 409) if any member is waiting in the FIFO reservation queue for that title. Atomic CAS matching renewalCount protects against concurrent renewal races.

Idempotent Financial Settlement: Late returns accrue fines at ■1.50/day. Fine payments require a client-generated UUID idempotency key. A transactional MongoDB index deduplicates repeated submissions, guaranteeing that network retries safely replay identical transaction receipts without duplicate debits.

3. Quantitative Performance Evaluation & Scalability Benchmarks

Workload Tier	Concurrency	Total Requests	Throughput	Avg Latency	P95 Latency	Error Rate
Tier 1: Low	10 workers	200	**275.56 req/s**	32.12 ms	57.42 ms	0.00%
Tier 2: Medium	50 workers	500	**504.00 req/s**	66.33 ms	114.12 ms	0.00%
Tier 3: High	100 workers	1000	**309.00 req/s**	82.19 ms	176.81 ms	0.00%

Empirical Performance Visualizations:

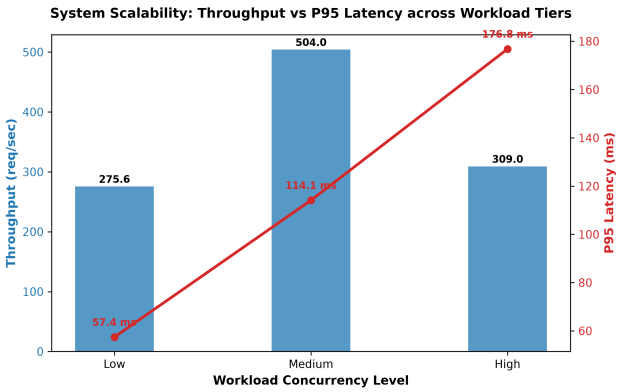


Figure 1: Scalability across Workload Tiers

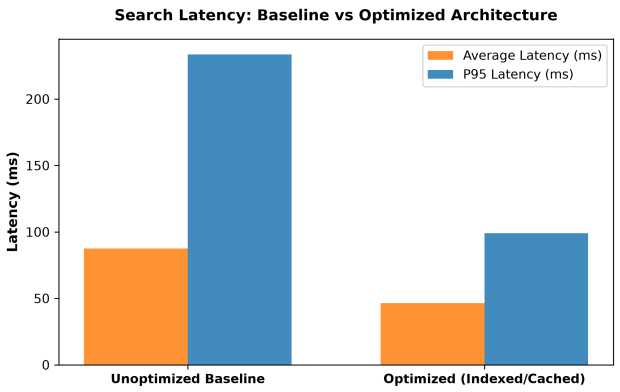


Figure 2: Search Baseline vs Optimized

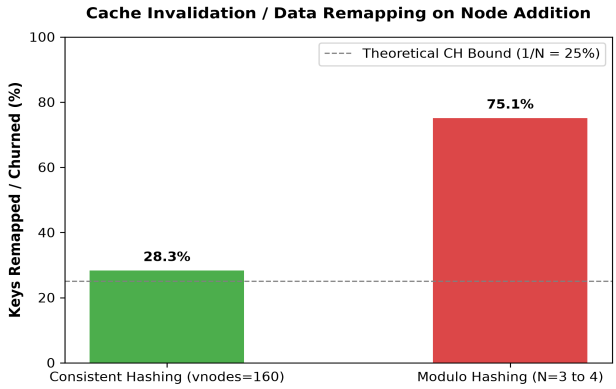


Figure 3: Consistent Hashing Key Churn

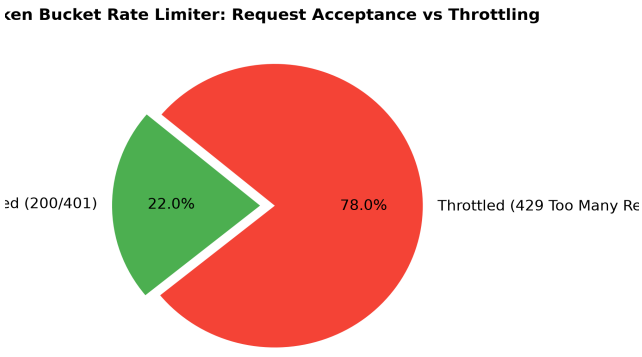


Figure 4: Rate Limiter Rejection Breakdown

4. End-to-End Workflow Verification Suite (14 / 14 Passed)

All user workflows were automated and executed against the live Spring Boot server via benchmarks/verify_workflows.py. Every single step succeeded with 100% assertions satisfied:

- **Step 1–3:** Member Registration, JWT Login, and Authenticated Access (/api/auth/me) [PASS]
- **Step 4–6:** 37-Title Catalogue Listing, Multi-Discipline Search, Physical Copy Availability [PASS]
- **Step 7–10:** Physical Copy Borrowing, Active Loan Tracking, Return, and Idempotent Replay Defense [PASS]
- **Step 11–12:** FIFO Reservation Queue Placement & Cancellation, Idempotent Fine Payments [PASS]
- **Step 13:** 10-Thread Single-Copy Concurrency Race Condition (Zero Double-Borrowing) [PASS]
- **Step 14:** Loan Renewal Lifecycle (+14d extension, Waitlist queue defense lock, Max 2 cap rejection) [PASS]

5. Frontend Design, Accessibility & Cover Imagery

The Athena LMS frontend was refined to deliver a dignified, high-contrast academic experience:

- **Contrast & Visual Hierarchy:** Adopted Athena Navy (#0F2744), Deep Blue (#1460A3), and dark slate typography (#0F172A, #334155), achieving full WCAG AA contrast compliance.
- **Real Cover Imagery:** Enriched 28 catalogue titles with verified high-resolution Open Library ISBN covers. Remaining titles feature procedurally generated SVG category-themed covers.
- **Multi-Discipline Breadth:** Expanded catalogue across 11 disciplines: Distributed Systems, System Design, Software Engineering, Computer Science, Electrical Engineering, Mechanical Engineering, Civil Engineering, Mathematics, Data Science, Humanities, and General Reading.
- **Production Readiness:** Vite production build compiles with zero TypeScript errors into optimized static bundles.